Multicollinearity: Detection and Remedies

1 Where We Left Off

A Recurring Issue

- Lecture 4: $\mathrm{Var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$, which becomes large when $\mathbf{x}_j$ is close to a linear combination of the other regressors.
- Lecture 8: centering x_1 and x_2 reduced the correlation between $x_1 x_2$ and its components.
- Lecture 9, gasoline mileage: all five t-tests insignificant ($p > 0.2$), yet the overall $F_0 = 14.66$ ($p \approx 10^{-6}$). Horsepower and torque are correlated at 0.96.
- Lecture 9, delivery time: adjusting distance for cases left only $1 - 0.824^2 = 0.321$ of its variation, and $1/0.321 = 3.12$ was called the variance inflation factor.

This lecture gives the problem a name, measures it, and looks at what can be done about it.

Datasets in This Lecture

Data	MPV	n
Acetylene	Table 9.1	16
Webster, Gunst & Mason	Table 9.4	12
Gasoline mileage	Table B.3	30
Hald cement	Table B.21	13
Delivery time	Example 3.1	25
Windmill	Table 5.5	25

MPV = Montgomery, Peck & Vining, *Introduction to Linear Regression Analysis*, 5th ed.

The Datasets, Briefly

- **Acetylene**: percentage conversion of n-heptane to acetylene in 16 runs of a chemical process; x_1 = reactor temperature (°C), x_2 = ratio of H_2 to n-heptane (mole ratio), x_3 = contact time (s).
- **Webster, Gunst & Mason**: 12 observations on y and six regressors; $x_1 + x_2 + x_3 + x_4$ is close to 10 in every row.
- **Gasoline mileage**: miles per gallon of 32 automobiles (30 with complete data) and ten design variables: displacement, horsepower, torque, compression ratio, rear axle ratio, carburetor barrels, transmission speeds, length, width, weight.
- **Hald cement**: heat evolved (cal/g) by 13 cement samples during hardening; x_1–x_4 are the percentages of four compounds in the clinker.
- **Delivery time**: vending-machine delivery time against cases stocked and distance walked.
- **Windmill**: DC output of a windmill against wind velocity.

Data Sources

- Acetylene: Kunugi, T., Tamura, T. and Naito, T. (1961), *Chemical Engineering Progress* 57(11); analysed by Marquardt, D. W. and Snee, R. D. (1975), Ridge regression in practice, *The American Statistician* 29(1), 3–20.
- Webster, J. T., Gunst, R. F. and Mason, R. L. (1974), Latent root regression analysis, *Technometrics* 16(4), 513–522.
- Gasoline mileage: *Motor Trend* (1975), as reproduced in MPV Table B.3.
- Hald cement: Woods, H., Steinour, H. H. and Starke, H. R. (1932), Effect of composition of Portland cement on heat evolved during hardening, *Industrial and Engineering Chemistry* 24(11), 1207–1214; Hald, A. (1952), *Statistical Theory with Engineering Applications*.
- Windmill: Joglekar, G., Schuenemeyer, J. H. and LaRiccia, V. (1989), Lack-of-fit testing when replicates are not available, *The American Statistician* 43(3), 135–143.
- Delivery time: MPV Example 3.1.

2 What Multicollinearity Is

Linear Dependence Among the Regressors

Write the regressor columns (without the intercept) as $\mathbf{x}_1, \ldots, \mathbf{x}_k$.

- **Exact dependence**: constants $t_1, \ldots, t_k$, not all zero, with

$$\sum_{j=1}^{k} t_j(\mathbf{x}_j - \bar{x}_j\mathbf{1}) = \mathbf{0}.$$

  Then $\mathbf{X}'\mathbf{X}$ is singular and $\boldsymbol{\beta}$ has no unique least-squares estimate.

- **Near dependence**: the same combination is close to $\mathbf{0}$. $\mathbf{X}'\mathbf{X}$ is invertible but nearly singular.

- **Multicollinearity** means a near dependence among the regressors. Exact dependence is rare in practice and is usually a coding error (for example, a dummy for every level of a factor).

Unit-Length Scaling

To compare regressors measured in different units, center and scale each column:

$$w_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{S_{jj}}}, \qquad S_{jj} = \sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2.$$

- Each column $\mathbf{w}_j$ has mean 0 and length 1: $\mathbf{w}_j'\mathbf{w}_j = 1$.

- $\mathbf{w}_j'\mathbf{w}_l = r_{jl}$, the correlation between regressors x_j and x_l, so $\mathbf{W}'\mathbf{W}$ is the **correlation matrix** of the regressors.

- Scaling y the same way gives the model $\mathbf{y}^* = \mathbf{W}\mathbf{b} + \boldsymbol{\varepsilon}^*$ with no intercept. The fitted coefficients $\hat{b}_j = \hat{\beta}_j\sqrt{S_{jj}/\mathrm{SS_T}}$ are unit-free.

- The errors are scaled too: $\boldsymbol{\varepsilon}^* = \boldsymbol{\varepsilon}/\sqrt{\mathrm{SS_T}}$ has variance $\sigma_w^2 = \sigma^2/\mathrm{SS_T}$.

All diagnostics in this lecture use $\mathbf{W}'\mathbf{W}$, so they do not depend on the units of the regressors.

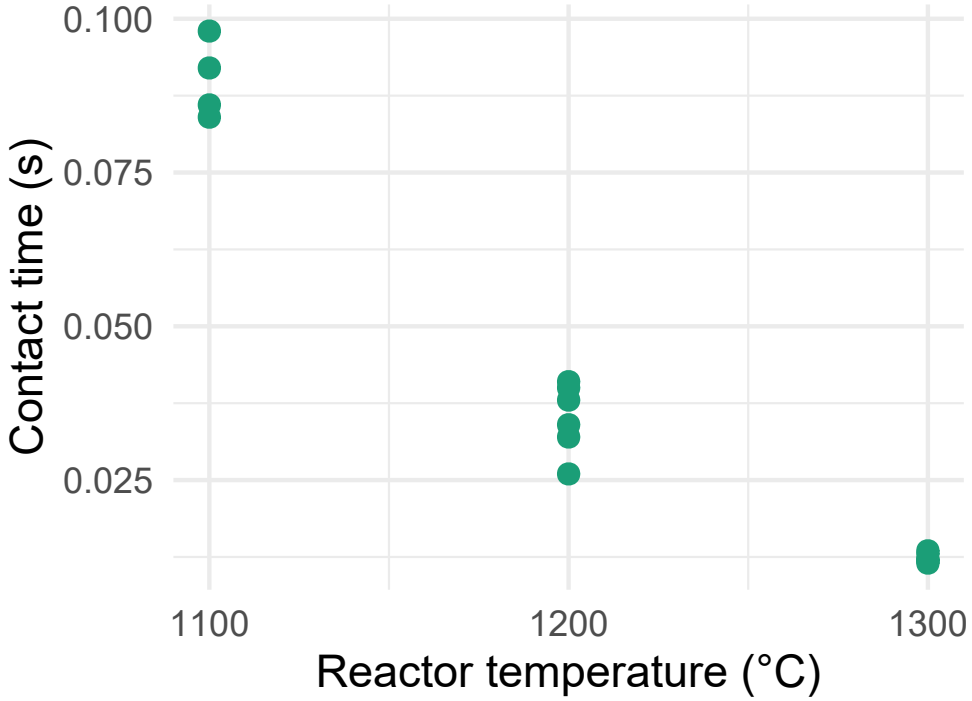
Sources of Multicollinearity

1. **Data collection**: only a subspace of the regressor space was sampled. Acetylene: temperature and contact time were changed together.
2. **Constraints in the model or population**: Hald cement, where the four compound percentages add up to 95–99; Webster data, where $x_1 + x_2 + x_3 + x_4 \approx 10$.
3. **Model specification**: polynomial or interaction terms. Windmill: over the observed range of wind velocity, x and x^2 are almost collinear.
4. **Overdefined model**: more regressors than observations, or nearly so, as in some medical and genetic studies.

The first source can be fixed by collecting more data. The second cannot: the constraint is a property of the population.

Acetylene Data: How the Data Were Collected

- 16 runs at three temperatures: 1100, 1200 and 1300°C.
- Contact time is long at low temperature and short at high temperature.
- $r(x_1, x_3) = -0.958$.
- No runs at high temperature with long contact time, or at low temperature with short contact time.



Acetylene Data: The Model

MPV fits a full quadratic model in the three regressors, after scaling each to mean 0 and standard deviation 1:

$$y = \beta_0 + \sum_{j=1}^{3} \beta_j x_j + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \sum_{j=1}^{3} \beta_{jj} x_j^2 + \varepsilon$$

- $k = 9$ regressors, $p = 10$ parameters, $n = 16$: only 6 residual degrees of freedom.
- Least squares: $R^2 = 0.9977$, $\mathrm{MS}_{\mathrm{Res}} = 0.8126$.
- Since $x_1 \approx -x_3$ in the scaled data, products of x_1 resemble products of x_3: $x_1 x_3 \approx -x_1^2$ and $x_1 x_2 \approx -x_2 x_3$. The quadratic terms add new near dependencies.

Question: Is Multicollinearity a Property of the Model or of the Data?

- The acetylene runs could, in principle, be repeated at other operating conditions. The cement percentages cannot be made to sum to anything other than about 100.
- In which case would more data remove the problem?

3 Effects of Multicollinearity

Two Regressors

In unit-length form with two regressors,

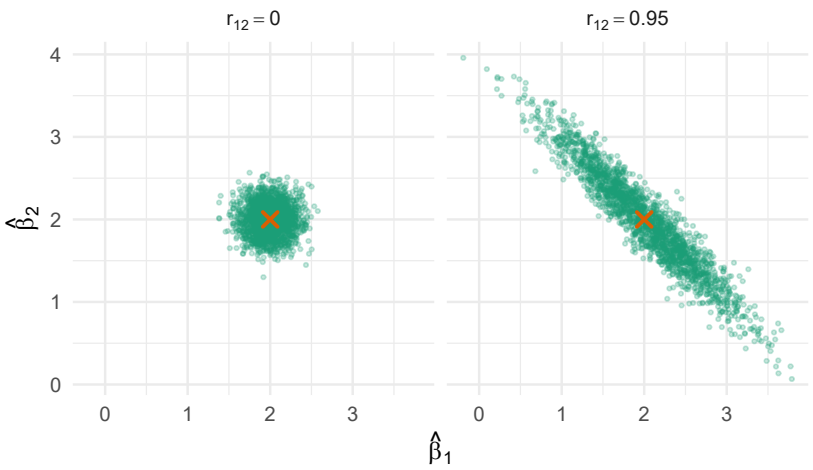
$$\mathbf{W}'\mathbf{W} = \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix}, \qquad \mathbf{C} = (\mathbf{W}'\mathbf{W})^{-1} = \frac{1}{1 - r_{12}^2} \begin{bmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{bmatrix}.$$

- $\mathrm{Var}(\hat{b}_1) = \mathrm{Var}(\hat{b}_2) = \sigma_w^2/(1 - r_{12}^2)$.
- $\mathrm{Cov}(\hat{b}_1, \hat{b}_2) = -r_{12}\, \sigma_w^2/(1 - r_{12}^2)$, so $\mathrm{Corr}(\hat{b}_1, \hat{b}_2) = -r_{12}$.
- As $|r_{12}| \to 1$ both variances grow without bound, and the estimates become strongly correlated, with the opposite sign to r_{12}.

From here on $\mathbf{C}$ means $(\mathbf{W}'\mathbf{W})^{-1}$, not Lecture 4's $(\mathbf{X}'\mathbf{X})^{-1}$. For a slope coefficient the two are related by $[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} = C_{jj}/S_{jj}$.

Simulation: Sampling Distribution of $(\hat{\beta}_1, \hat{\beta}_2)$

$y = 1 + 2x_1 + 2x_2 + \varepsilon$, $\sigma = 1$, $n = 30$, the same $\mathbf{X}$ in 2000 replicates; x_1, x_2 have sample standard deviation 1.



$r_{12} = 0$: $\mathrm{sd}(\hat{\beta}_1) = 0.18$. $r_{12} = 0.95$: $\mathrm{sd}(\hat{\beta}_1) = 0.60$ (theory 0.595) and $\mathrm{Corr}(\hat{\beta}_1, \hat{\beta}_2) = -0.95$.

Some Combinations Are Still Well Estimated

With $\mathbf{C}$ from the two-regressor case,

$$\mathrm{Var}(\hat{b}_1 + \hat{b}_2) = \sigma_w^2 \frac{2 - 2r_{12}}{1 - r_{12}^2} = \frac{2\sigma_w^2}{1 + r_{12}},$$
$$\mathrm{Var}(\hat{b}_1 - \hat{b}_2) = \sigma_w^2 \frac{2 + 2r_{12}}{1 - r_{12}^2} = \frac{2\sigma_w^2}{1 - r_{12}}.$$

- For $r_{12} = 0.95$: the sum has variance $1.03\sigma_w^2$, the difference $40\sigma_w^2$.
- In the scatter plot, the cloud stretches along the line $\hat{\beta}_1 + \hat{\beta}_2 = 4$ and is narrow across it.
- The data say a lot about the combined effect of x_1 and x_2, and little about how to split it between them.

The General Case

For k regressors in unit-length form, $\mathbf{C} = (\mathbf{W}'\mathbf{W})^{-1}$ and

$$C_{jj} = \frac{1}{1 - R_j^2}, \qquad \mathrm{Var}(\hat{b}_j) = \frac{\sigma_w^2}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination from regressing x_j on the other $k-1$ regressors (derived in the VIF section). Since $\hat{\beta}_j = \hat{b}_j \sqrt{\mathrm{SS_T}/S_{jj}}$ and $\sigma_w^2 = \sigma^2/\mathrm{SS_T}$, in original units

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{S_{jj}} \cdot \frac{1}{1 - R_j^2},$$

the same formula as in the added-variable plot of Lecture 9.

The Estimates Are Too Long on Average

Let $L^2 = (\widehat{\mathbf{b}} - \mathbf{b})'(\widehat{\mathbf{b}} - \mathbf{b})$ be the squared distance from the estimate to the true coefficients. With eigenvalues $\lambda_1, \ldots, \lambda_k$ of $\mathbf{W}'\mathbf{W}$,

$$\mathrm{E}(L^2) = \sum_{j=1}^{k} \mathrm{Var}(\hat{b}_j) = \sigma_w^2 \,\mathrm{tr}(\mathbf{W}'\mathbf{W})^{-1} = \sigma_w^2 \sum_{j=1}^{k} \frac{1}{\lambda_j},$$

$$\mathrm{E}(\widehat{\mathbf{b}}'\widehat{\mathbf{b}}) = \mathbf{b}'\mathbf{b} + \sigma_w^2 \sum_{j=1}^{k} \frac{1}{\lambda_j}.$$

- The trace of a matrix is the sum of its eigenvalues, and the eigenvalues of $(\mathbf{W}'\mathbf{W})^{-1}$ are $1/\lambda_j$.
- One small λ_j is enough to make $\mathrm{E}(L^2)$ large. Orthogonal regressors have all $\lambda_j = 1$ and $\mathrm{E}(L^2) = k\sigma_w^2$.

Acetylene Data: Length of the Estimate

- $\sum_j 1/\lambda_j = 10{,}610$, against 9 for orthogonal regressors.
- In the unit-length scale, $\hat{\sigma}_w^2 = \mathrm{MS}_{\mathrm{Res}}/\mathrm{SS}_{\mathrm{T}} = 0.8126/2123.7 = 0.000383$.
- Least squares: $\widehat{\mathbf{b}}'\widehat{\mathbf{b}} = 6.77$.
- Expected inflation from noise alone: $\hat{\sigma}_w^2 \sum_j 1/\lambda_j = 0.000383 \times 10{,}610 = 4.06$.

More than half of the squared length of the least-squares coefficient vector can be accounted for by the variance, not by the true coefficients.

Symptoms in a Fitted Model

- Large standard errors and wide confidence intervals.
- Individual t-tests insignificant while the overall F-test is highly significant.
- Coefficients with the wrong sign or an implausible size.
- Estimates that change a lot when an observation or a regressor is added or removed.

Gasoline mileage, all ten regressors: $R^2 = 0.829$ and $F_0 = 9.18$ on $(10, 19)$ df, yet no t-test is significant at 5%. Displacement gets -0.100 but torque gets $+0.126$, although both are correlated about -0.85 with mileage.

Prediction Is Not Always Hurt

$$\mathrm{Var}(\hat{y}_0) = \sigma^2 \,\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

- If $\mathbf{x}_0$ follows the same pattern as the data (acetylene: high temperature with short contact time), $\hat{y}_0$ is precise: the errors in the correlated $\hat{\beta}_j$ cancel through their covariances (next slide).
- If $\mathbf{x}_0$ breaks the pattern (high temperature with long contact time), the prediction depends on exactly those directions the data cannot estimate, and $\mathrm{Var}(\hat{y}_0)$ is large.
- Multicollinearity is mainly a problem for *interpreting* coefficients and for *extrapolating* outside the region of the data.

Why the Errors Cancel: Two Regressors

In unit-length form, a new point (w_{01}, w_{02}) has

$$\mathrm{Var}(w_{01}\hat{b}_1 + w_{02}\hat{b}_2) = \sigma_w^2 \, \frac{w_{01}^2 + w_{02}^2 - 2r_{12}\, w_{01} w_{02}}{1 - r_{12}^2}.$$

	$r_{12} > 0$	$r_{12} < 0$ (acetylene)
Data pattern	rise together	one up, one down
$\mathrm{Cov}(\hat{b}_1, \hat{b}_2)$	< 0: errors opposite	> 0: errors alike
Point along the pattern	$w_{02} = w_{01}$	$w_{02} = -w_{01}$
Weights on the errors	same sign	opposite signs
Variance along the pattern (small)	$2w_{01}^2\sigma_w^2/(1 + \lvert r_{12}\rvert)$	$2w_{01}^2\sigma_w^2/(1 + \lvert r_{12}\rvert)$
Variance against it (large)	$2w_{01}^2\sigma_w^2/(1 - \lvert r_{12}\rvert)$	$2w_{01}^2\sigma_w^2/(1 - \lvert r_{12}\rvert)$

Whatever the sign of r_{12}, the errors cancel when the new point lies along the pattern of the data.

Question: Does Multicollinearity Bias Least Squares?

- The Gauss–Markov conditions still hold. Is $\hat{\boldsymbol{\beta}}$ still unbiased? Still BLUE?
- If so, what exactly is the problem?

4 Detection: The Correlation Matrix

Inspecting W′W

The off-diagonal elements r_{jl} of $\mathbf{W}'\mathbf{W}$ are the pairwise correlations between regressors.

- $|r_{jl}|$ close to 1 signals a near dependence between x_j and x_l.
- Easy to compute and to read, and it shows *which* pair is involved.

Acetylene, original regressors: $r(x_1, x_3) = -0.958$, $r(x_1, x_2) = 0.224$, $r(x_2, x_3) = -0.240$. Among the nine terms of the quadratic model, the largest correlations are

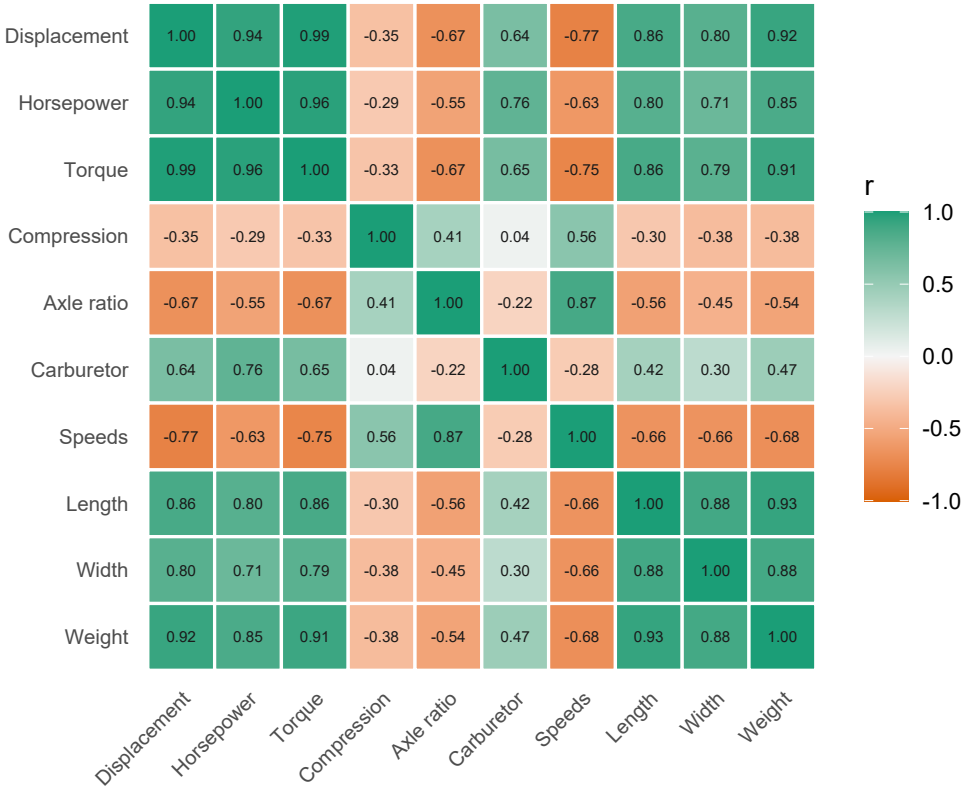
Pair	r	Pair	r
x_1x_2, x_2x_3	-0.974	x_1, x_3	-0.958
x_1x_3, x_1^2	-0.972	x_1^2, x_3^2	0.894
x_1x_3, x_3^2	-0.972	x_3, x_3^2	0.765

Gasoline Mileage: Correlation Matrix

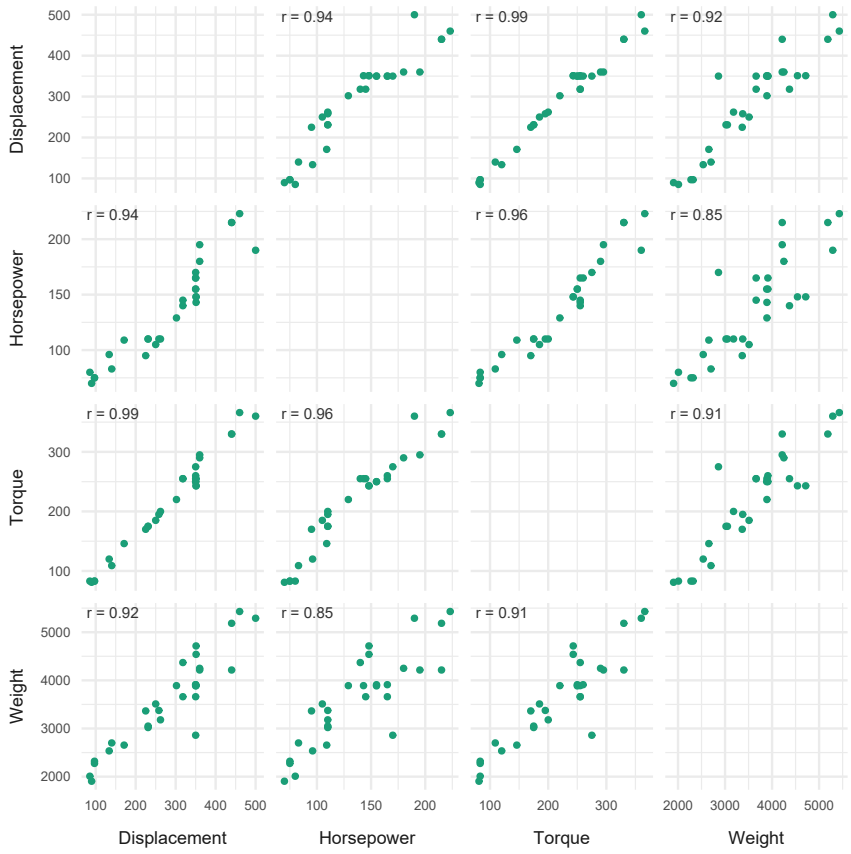
Six pairs exceed 0.9:

- displacement–torque 0.990
- horsepower–torque 0.964
- displacement–horsepower 0.941
- length–weight 0.934
- displacement–weight 0.924
- torque–weight 0.910

Displacement, horsepower, torque and weight all measure the size of the car.



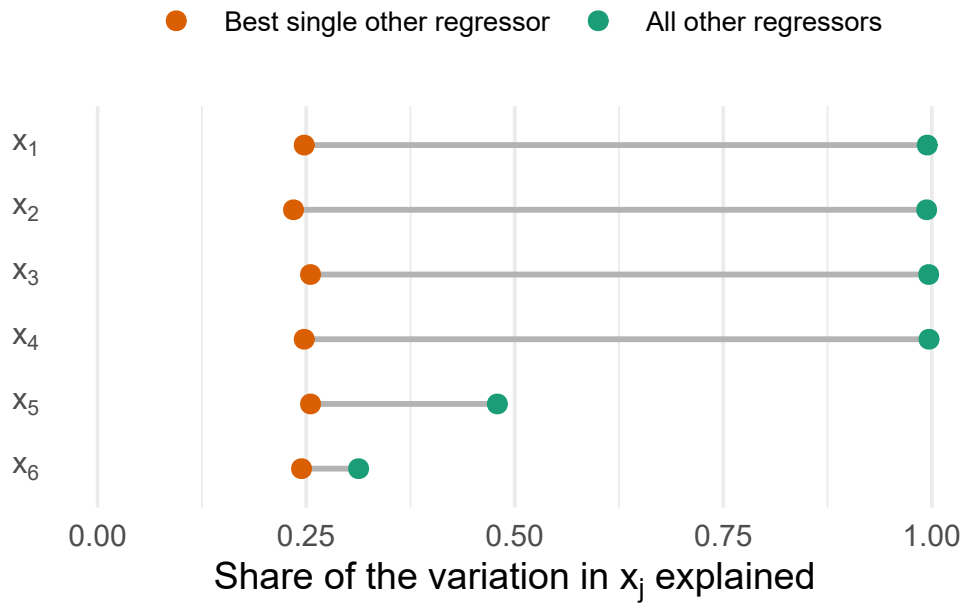
Gasoline Mileage: The Four Size Variables



Limitation: Only Pairs

A near dependence can involve three or more regressors without any large pairwise correlation. Webster, Gunst & Mason data:

- Largest $|r_{jl}|$ is 0.505.
- Yet $x_1 + x_2 + x_3 + x_4 = 10$ in 11 of the 12 rows (11 in the first).
- The best single other regressor explains at most 26% of the variation in any x_j.
- All the others together explain at least 99.4% of each of $x_1, \ldots, x_4$.



Question: Is a Small Pairwise Correlation Reassuring?

- With exactly two regressors, does a small $|r_{12}|$ rule out multicollinearity?
- With $k \geq 3$ regressors, does it?

5 Detection: Variance Inflation Factors

Variance Inflation Factor

$$\text{VIF}_j = C_{jj} = \left[(\mathbf{W}'\mathbf{W})^{-1}\right]_{jj} = \frac{1}{1 - R_j^2}$$

- R_j^2: coefficient of determination from regressing x_j on the other regressors.
- $\text{Var}(\hat{\beta}_j) = \dfrac{\sigma^2}{S_{jj}} \text{VIF}_j$: the factor by which the variance is inflated compared with regressors that are uncorrelated with x_j.
- Standard errors and confidence intervals are inflated by $\sqrt{\text{VIF}_j}$.
- $\text{VIF}_j \geq 1$, with equality when x_j is uncorrelated with every other regressor.

Why $C_{jj} = 1/(1 - R_j^2)$

Put $\mathbf{w}_j$ first: $\mathbf{W} = [\mathbf{w}_j \;\; \mathbf{W}_{(-j)}]$. By the formula for the inverse of a partitioned matrix,

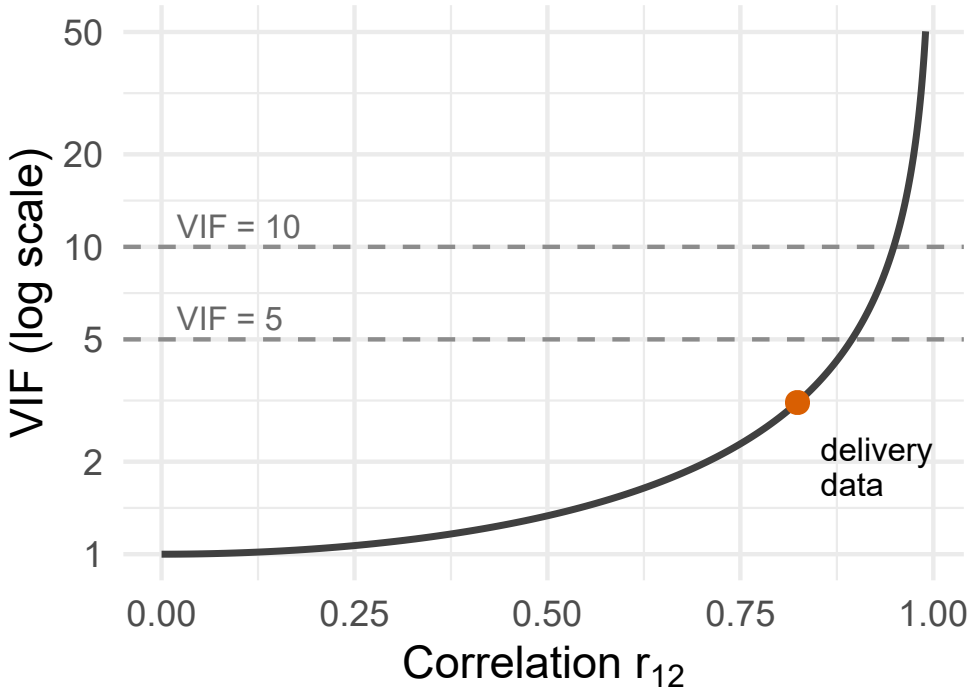
$$C_{jj} = \left[\mathbf{w}_j'\mathbf{w}_j - \mathbf{w}_j'\mathbf{W}_{(-j)}\left(\mathbf{W}_{(-j)}'\mathbf{W}_{(-j)}\right)^{-1}\mathbf{W}_{(-j)}'\mathbf{w}_j\right]^{-1} = \left[\mathbf{w}_j'(\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{w}_j\right]^{-1}.$$

- $\mathbf{w}_j'(\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{w}_j$ is the residual sum of squares from regressing $\mathbf{w}_j$ on $\mathbf{W}_{(-j)}$.

- $\mathbf{w}_j$ has mean 0 and $\mathbf{w}_j'\mathbf{w}_j = 1$, so its total sum of squares is 1 and this residual sum of squares equals $1 - R_j^2$.
- The same quantity appeared in Lecture 9 as the horizontal spread of the added-variable plot.

How Large Is Too Large?

- Common guideline: $\text{VIF}_j > 5$ or $\text{VIF}_j > 10$ indicates a poorly estimated coefficient.
- VIF $= 10$ means $R_j^2 = 0.9$ and standard errors $\sqrt{10} = 3.2$ times larger.
- With two regressors, VIF $= 1/(1 - r_{12}^2)$; the delivery data ($r = 0.824$) sit at 3.12.



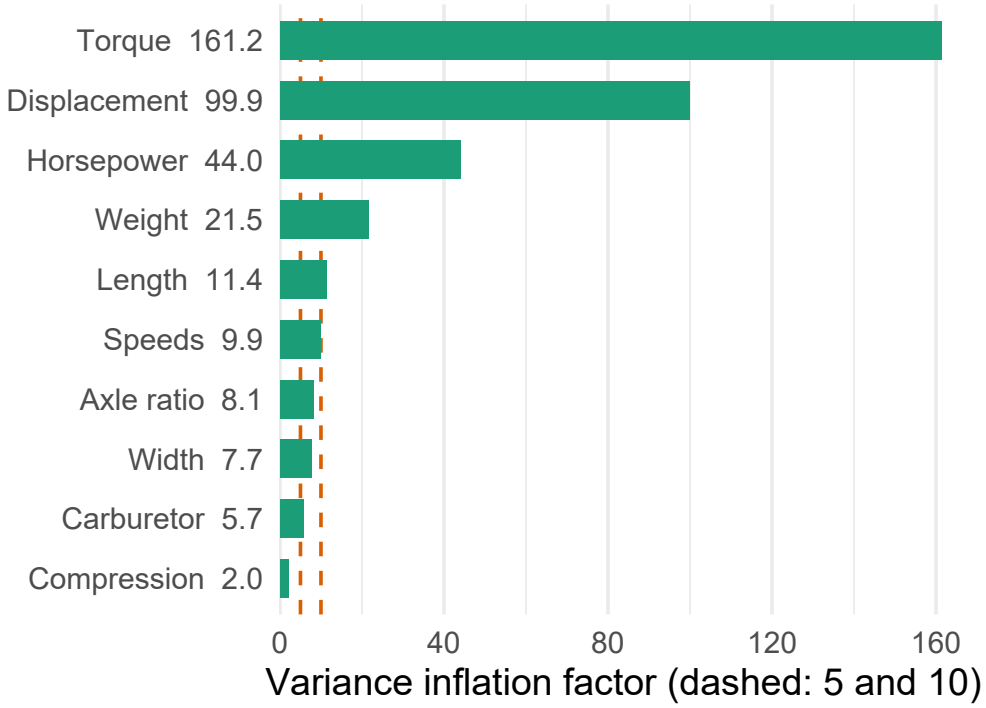
Acetylene Data: VIFs

Term	VIF	R_j^2	Term	VIF	R_j^2
x_1	375.25	0.9973	x_2x_3	35.61	0.9719
x_2	1.74	0.4255	x_1^2	1762.58	0.9994
x_3	680.28	0.9985	x_2^2	3.16	0.6840
x_1x_2	31.04	0.9678	x_3^2	1156.77	0.9991
x_1x_3	6563.35	0.9998			

- Every term involving temperature or contact time has a VIF above 30; the hydrogen ratio terms x_2 and x_2^2 are fine.
- The standard error of $\hat{\beta}_{13}$ is $\sqrt{6563} = 81$ times what it would be with orthogonal regressors.

Gasoline Mileage: VIFs for All Ten Regressors

- Five VIFs exceed 10: torque, displacement, horsepower, weight, length.
- Torque: $R_j^2 = 0.994$. The other nine regressors predict torque almost perfectly.
- Compression ratio is nearly unaffected.



Back to the Lecture 9 Puzzle

The five-regressor gasoline model: every t-test insignificant, $F_0 = 14.66$.

Regressor	VIF	$\sqrt{\text{VIF}}$	p-value
Horsepower	29.05	5.4	0.882
Torque	47.34	6.9	0.676
Rear axle ratio	3.18	1.8	0.332
Carburetor barrels	3.80	1.9	0.971
Weight	8.98	3.0	0.205

- The standard errors of horsepower and torque are five to seven times what they would be without collinearity.
- Each t-test asks whether x_j helps *given the others*. When the others carry the same information, the answer is no, even though the group as a whole matters.

Webster and Cement Data: VIFs

	x_1	x_2	x_3	x_4	x_5	x_6
Webster	182.05	161.36	266.26	297.71	1.92	1.46
Hald cement	38.50	254.42	46.87	282.51		

- Webster: the VIFs flag x_1–x_4 even though no pairwise correlation exceeds 0.51.
- Cement: $R^2 = 0.982$ and the overall F-test is highly significant, yet no coefficient is significant at 5%.

VIFs in R, and What They Do Not Tell

- `car::vif(fit)`, or directly `diag(solve(cor(X)))` for a numeric regressor matrix `X`.
- VIFs say *which* coefficients are poorly estimated.
- They do not say how many near dependencies there are, or which regressors take part in each. Acetylene: seven terms have VIF above 30. Is that one dependency or several?

The eigenvalues and eigenvectors of $\mathbf{W}'\mathbf{W}$ answer these questions.

Question: Can a VIF Be Less Than 1?

- Could adding a regressor correlated with x_j ever make $\hat{\beta}_j$ *more* precise?

6 Detection: Eigensystem Analysis

Eigenvalues of $\mathbf{W}'\mathbf{W}$

$$\mathbf{W}'\mathbf{W} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}', \qquad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k), \quad \lambda_1 \geq \cdots \geq \lambda_k > 0$$

- $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_k]$ is orthogonal: the eigenvectors are orthonormal.
- $\sum_j \lambda_j = \text{tr}(\mathbf{W}'\mathbf{W}) = k$, since each diagonal element is 1.
- Orthogonal regressors: all $\lambda_j = 1$.
- A near dependence shows up as a small eigenvalue:

$$(\mathbf{W}\mathbf{v}_j)'(\mathbf{W}\mathbf{v}_j) = \mathbf{v}_j'\mathbf{W}'\mathbf{W}\mathbf{v}_j = \lambda_j.$$

If $\lambda_j \approx 0$, the combination $\mathbf{W}\mathbf{v}_j = \sum_l v_{lj}\mathbf{w}_l$ is close to $\mathbf{0}$ for every observation.

Condition Number and Condition Indices

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}, \qquad \kappa_j = \frac{\lambda_{\max}}{\lambda_j}, \ j = 1, \dots, k$$

Condition number	Multicollinearity
$\kappa < 100$	no serious problem
$100 \leq \kappa \leq 1000$	moderate to strong
$\kappa > 1000$	severe

- The number of condition indices κ_j above 1000 counts the near dependencies that are severe.
- $\det(\mathbf{W}'\mathbf{W}) = \prod_j \lambda_j$ lies between 0 and 1 and is close to 0 under multicollinearity, but it does not say how many dependencies there are.

Acetylene Data: Eigenvalues

j	λ_j	κ_j	j	λ_j	κ_j
1	4.2052	1.0	6	0.0495	84.9
2	2.1620	1.9	7	0.0136	308.6
3	1.1387	3.7	8	0.0051	820.1
4	1.0405	4.0	9	0.000097	43,381
5	0.3852	10.9			

- $\kappa = 43{,}381$: severe. One condition index exceeds 1000, two more lie between 100 and 1000.
- $\det(\mathbf{W}'\mathbf{W}) = 1.4 \times 10^{-9}$.
- The eigenvalues sum to $9 = k$, but four of them hold almost all of it.

Acetylene Data: Which Terms Are Involved?

Eigenvector for $\lambda_9 = 0.000097$:

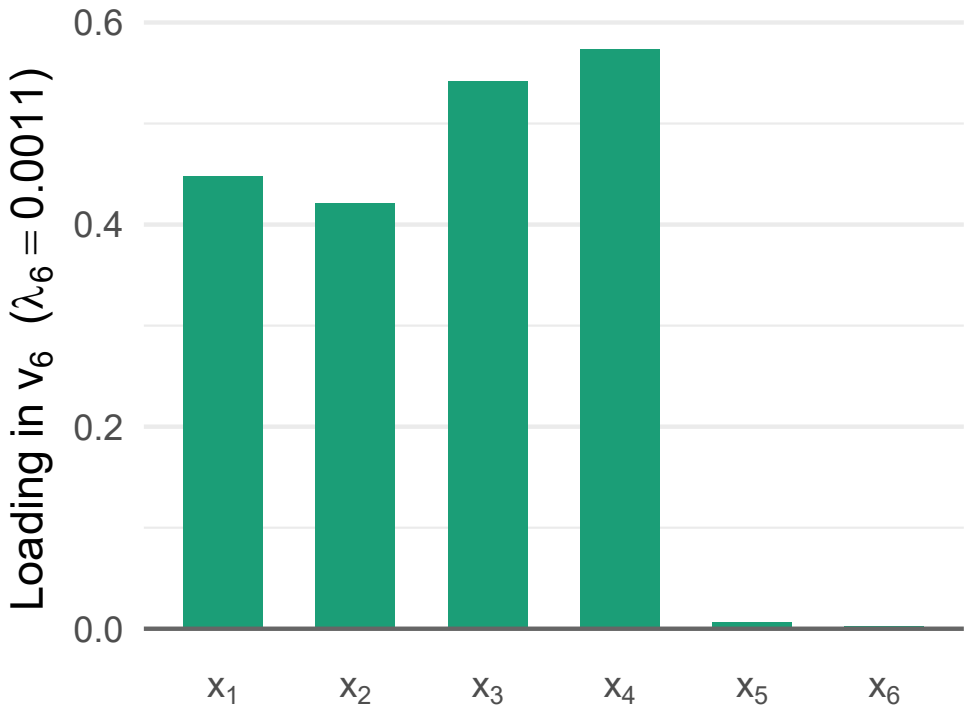
x_1	x_2	x_3	x_1x_2	x_1x_3	x_2x_3	x_1^2	x_2^2	x_3^2
0.174	−0.003	0.237	0.003	0.797	0.008	0.412	0.005	0.329

$$0.17\,\mathbf{w}_{x_1} + 0.24\,\mathbf{w}_{x_3} + 0.80\,\mathbf{w}_{x_1x_3} + 0.41\,\mathbf{w}_{x_1^2} + 0.33\,\mathbf{w}_{x_3^2} \approx \mathbf{0}$$

- Only temperature and contact time terms take part; the hydrogen ratio does not.
- The largest element of $\mathbf{W}\mathbf{v}_9$ is 0.0045 in absolute value.

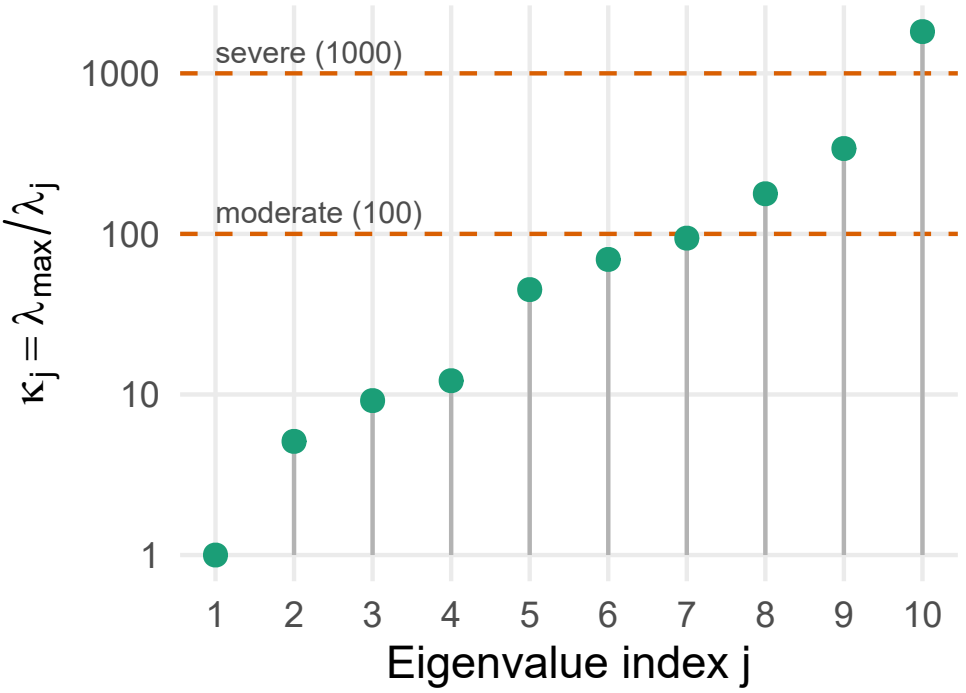
Webster Data: One Hidden Dependency

- λ: 2.4288, 1.5462, 0.9221, 0.7940, 0.3079, 0.0011.
- $\kappa = 2196$: severe, one near dependency.
- $\mathbf{v}_6$ loads only on x_1–x_4.
- Dividing the loadings by the column lengths $\sqrt{S_{jj}}$ gives 1.00, 1.06, 1.05, 1.05, 0.04, 0.03: in original units the dependency is $x_1 + x_2 + x_3 + x_4 \approx$ constant.



Gasoline Mileage: Condition Indices

- $\kappa = 1820$: severe.
- One κ_j above 1000, two between 100 and 1000.
- Eigenvector of the smallest eigenvalue: displacement 0.55, horsepower 0.31, torque -0.77; all others below 0.08 in absolute value.
- The worst dependency links displacement, horsepower and torque.



Variance-Decomposition Proportions

Belsley, Kuh and Welsch (1980) split each variance by eigenvalue. From $\mathbf{C} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}'$,

$$\text{VIF}_j = C_{jj} = \sum_{s=1}^{k} \frac{v_{js}^2}{\lambda_s}, \qquad \pi_{sj} = \frac{v_{js}^2/\lambda_s}{\text{VIF}_j}.$$

- π_{sj} is the share of $\text{Var}(\hat{b}_j)$ that comes from eigenvalue s; each column of proportions sums to 1.
- Condition index $\eta_s = \sqrt{\lambda_{\max}/\lambda_s}$ (the square root of κ_s).
- Rule: a row with $\eta_s > 30$ and two or more $\pi_{sj} > 0.5$ marks a dependency among those regressors.
- Belsley, Kuh and Welsch use $\mathbf{X}$ scaled to unit length but not centered, with the intercept column; here we use $\mathbf{W}$ to match the rest of the lecture.

Acetylene Data: Variance-Decomposition Proportions

s	η_s	x_1	x_2	x_3	x_1x_2	x_1x_3	x_2x_3	x_1^2	x_2^2	x_3^2
5	3.3	.000	.508	.000	.002	.000	.003	.000	.442	.000
6	9.2	.003	.005	.000	.087	.000	.056	.003	.032	.007
7	17.6	.010	.005	.003	**.822**	.000	**.793**	.004	.000	.005
8	28.6	.151	.093	.146	.077	.001	.121	.000	.353	.023
9	208.3	**.835**	.065	**.851**	.002	**.999**	.020	**.992**	.082	**.965**

Rows 1–4 are omitted (all proportions small except x_2: 0.290 in row 4).

- Row 9: the temperature and contact time terms, as the eigenvector showed.
- Row 7 ($\eta = 17.6$, below 30 but clear): x_1x_2 and x_2x_3, which are nearly opposite because $x_1 \approx -x_3$.

Question: Why Must Some Eigenvalue Be Small?

- The eigenvalues of $\mathbf{W}'\mathbf{W}$ always add up to k. If $\lambda_1 = 4.2$ for the acetylene data, what does that say about the others?
- Is a large λ_1 by itself a sign of multicollinearity?

7 Remedies

Three Kinds of Remedy

1. **Collect more data** so that the dependency is broken.
2. **Respecify the model**: center, drop or combine regressors.
3. **Use a biased estimator** with much smaller variance: ridge regression, principal-component regression.

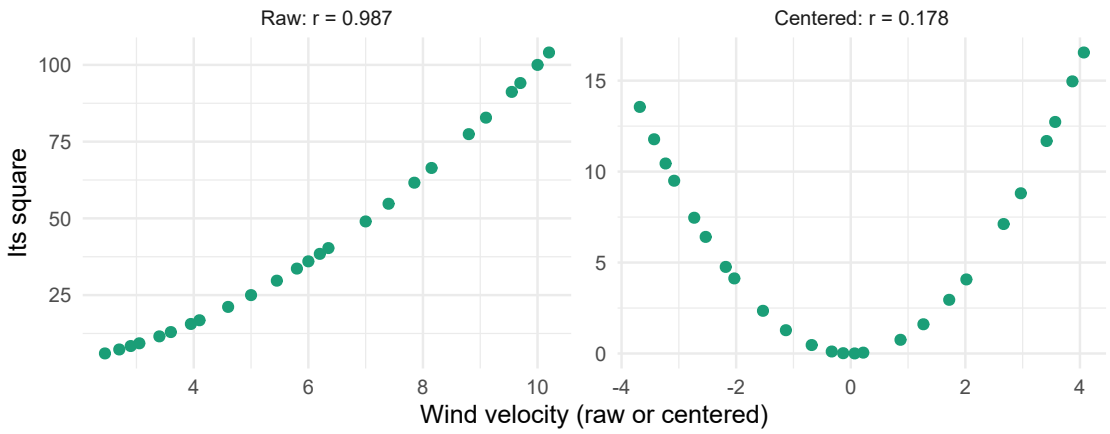
Which one fits depends on the source of the problem and on whether the goal is interpreting coefficients or predicting.

Collecting Additional Data

- New observations should fill the empty part of the regressor space. Acetylene: runs at high temperature with long contact time and at low temperature with short contact time.
- This is the best remedy when it is possible, because it removes the cause rather than working around it.
- Often it is not possible:
  - the process may not run, or may be unsafe, at the missing combinations;
  - observational data cannot be designed;
  - a constraint in the population (cement percentages) holds in every new sample too.

Respecification I: Centering Polynomial Terms

Windmill data, quadratic in wind velocity x (range 2.45 to 10.2, mean 6.13):



VIF falls from 38.5 to 1.03 when x is replaced by $x - \bar{x}$.

Centering Changes the Parameterization, Not the Fit

	Raw x	Centered $x - \bar{x}$
Intercept	-1.156	1.844
Linear (se)	0.7229 (0.0614)	0.2554 (0.0101)
Quadratic (se)	-0.0381 (0.0048)	-0.0381 (0.0048)

- Same fitted values, same R^2, same quadratic coefficient.
- The linear coefficient now means the slope at $x = \bar{x}$, which the data estimate well.
- Why it works: for $u = x - \bar{x}$, $\sum_i u_i \cdot u_i^2 = \sum_i u_i^3$, which is close to 0 when the u_i are spread symmetrically around 0. Far from 0, x^2 is almost a straight-line function of x.
- Centering does not help when the dependency is between different regressors: $\mathbf{W}$ is already centered.

Respecification II: Dropping or Combining Regressors

Hald cement: drop x_4, which takes part in the near-sum constraint.

	x_1	x_2	x_3
Estimate (se)	1.696 (0.205)	0.657 (0.044)	0.250 (0.185)
VIF	3.25	1.06	3.14

- R^2 goes from 0.9824 to 0.9823: x_4 added almost nothing that x_1–x_3 did not already carry.
- Dropping a regressor whose true coefficient is not zero biases the others. Which regressors to keep is the subject of the next lecture.
- Combining: replace highly correlated regressors by one index (for example, a single measure of engine size in place of displacement, horsepower and torque).

Ridge Regression

In unit-length form, for a constant $\theta \geq 0$,

$$\hat{\mathbf{b}}_R = (\mathbf{W}'\mathbf{W} + \theta\mathbf{I})^{-1}\mathbf{W}'\mathbf{y}^*.$$

- $\theta = 0$ gives least squares.
- Adding $\theta\mathbf{I}$ adds θ to every eigenvalue: $\lambda_j \to \lambda_j + \theta$. The smallest eigenvalues, which caused the trouble, change the most in relative terms.
- Equivalently, $\hat{\mathbf{b}}_R$ minimizes the penalized residual sum of squares

$$(\mathbf{y}^* - \mathbf{W}\mathbf{b})'(\mathbf{y}^* - \mathbf{W}\mathbf{b}) + \theta\,\mathbf{b}'\mathbf{b}.$$

  Setting the gradient $-2\mathbf{W}'\mathbf{y}^* + 2\mathbf{W}'\mathbf{W}\mathbf{b} + 2\theta\mathbf{b}$ to zero gives $\hat{\mathbf{b}}_R$. Large coefficients are penalized, so $\hat{\mathbf{b}}_R$ is shorter than $\hat{\mathbf{b}}$.

Ridge as Least Squares on Augmented Data

Append k pseudo-observations: rows $\sqrt{\theta}\,\mathbf{I}$ to $\mathbf{W}$ and zeros to $\mathbf{y}^*$:

$$\mathbf{W}_A = \begin{bmatrix} \mathbf{W} \\ \sqrt{\theta}\,\mathbf{I} \end{bmatrix}, \qquad \mathbf{y}_A = \begin{bmatrix} \mathbf{y}^* \\ \mathbf{0} \end{bmatrix}.$$

- $\mathbf{W}_A'\mathbf{W}_A = \mathbf{W}'\mathbf{W} + \theta\mathbf{I}$ and $\mathbf{W}_A'\mathbf{y}_A = \mathbf{W}'\mathbf{y}^*$, so least squares on the augmented data gives $\hat{\mathbf{b}}_R$.
- The pseudo-observations fill in the missing directions of the regressor space, with responses that pull the coefficients toward 0.
- Ridge regression can be fit with any least-squares routine.

Bias and Variance of the Ridge Estimator

$\hat{\mathbf{b}}_R = \mathbf{Z}_\theta\hat{\mathbf{b}}$ with $\mathbf{Z}_\theta = (\mathbf{W}'\mathbf{W} + \theta\mathbf{I})^{-1}\mathbf{W}'\mathbf{W}$, so

$$\begin{aligned} \mathrm{E}(\hat{\mathbf{b}}_R) &= \mathbf{Z}_\theta\mathbf{b} \neq \mathbf{b} \quad (\theta > 0), \\ \mathrm{Var}(\hat{\mathbf{b}}_R) &= \sigma_w^2(\mathbf{W}'\mathbf{W} + \theta\mathbf{I})^{-1}\mathbf{W}'\mathbf{W}(\mathbf{W}'\mathbf{W} + \theta\mathbf{I})^{-1}. \end{aligned}$$

With $\boldsymbol{\alpha} = \mathbf{V}'\mathbf{b}$ (coefficients along the eigenvectors), the mean squared error is

$$\mathrm{E}[(\hat{\mathbf{b}}_R - \mathbf{b})'(\hat{\mathbf{b}}_R - \mathbf{b})] = \underbrace{\sigma_w^2 \sum_{j=1}^{k} \frac{\lambda_j}{(\lambda_j + \theta)^2}}_{\text{total variance}} + \underbrace{\theta^2 \sum_{j=1}^{k} \frac{\alpha_j^2}{(\lambda_j + \theta)^2}}_{\text{squared bias}}.$$

- The variance falls and the squared bias rises as θ increases.
- Hoerl and Kennard (1970): there is always some $\theta > 0$ for which the ridge MSE is smaller than the least-squares MSE $\sigma_w^2 \sum_j 1/\lambda_j$.

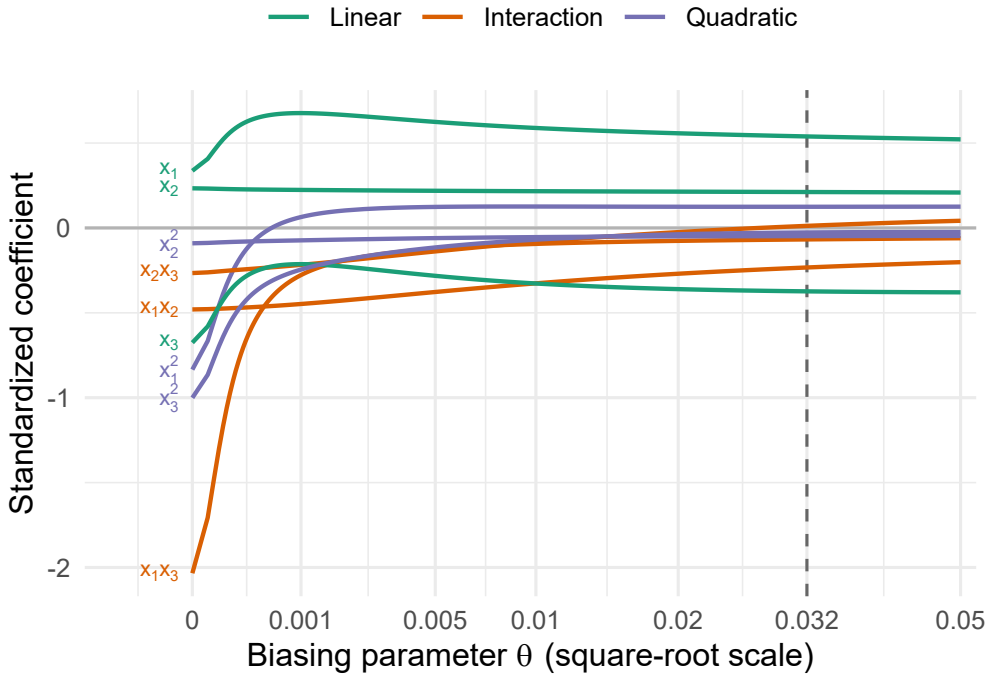
Choosing θ

- **Ridge trace**: plot each $\hat{b}_{R,j}$ against θ. Choose the smallest θ at which the coefficients have stabilized, have sensible signs, and SS_{Res} has not grown much.
- **VIFs of the ridge estimator**: the diagonal of $(\mathbf{W}'\mathbf{W}+\theta\mathbf{I})^{-1}\mathbf{W}'\mathbf{W}(\mathbf{W}'\mathbf{W}+\theta\mathbf{I})^{-1}$; choose θ so that they are all small (they can fall below 1).
- **Hoerl, Kennard and Baldwin (1975)**: $\theta = k\hat{\sigma}_w^2/\hat{\mathbf{b}}'\hat{\mathbf{b}}$, with $\hat{\sigma}_w^2$ and $\hat{\mathbf{b}}$ from least squares in the unit-length scale.
- **Cross-validation** or generalized cross-validation: choose θ to minimize prediction error.

The choice is a judgment call, and the methods often disagree.

Acetylene Data: Ridge Trace

- x_1x_3, x_1^2 and x_3^2 change most: the coefficient of x_1x_3 goes from -2.03 to -0.19 by $\theta = 0.002$.
- The trace has flattened by $\theta \approx 0.03$.
- We take $\theta = 0.032$ (dashed line), the value MPV chooses.
- Square-root scale for θ: most of the change happens at very small θ.



Acetylene Data: Fit and VIFs Along the Trace

θ	0	0.001	0.005	0.01	0.02	0.032	0.05
R^2	0.9977	0.9972	0.9967	0.9961	0.9952	0.9944	0.9933
Largest VIF	6563	76.0	27.0	12.4	6.39	4.18	2.63

- A tiny θ removes most of the variance inflation at almost no cost in fit.
- At $\theta = 0.032$: $\mathrm{SS}_{\mathrm{Res}}$ grows from 4.88 to 11.92, R^2 drops by 0.003, and every VIF is below 4.2.
- The Hoerl–Kennard–Baldwin rule gives $\theta = 0.0005$, which leaves the largest VIF above 170. Here it shrinks far less than the trace suggests.

Acetylene Data: Least Squares vs. Ridge

Coefficients in the unit-length scale:

Term	LS	Ridge	Term	LS	Ridge
x_1	0.337	0.539	x_2x_3	−0.266	0.012
x_2	0.234	0.212	x_1^2	−0.835	0.125
x_3	−0.676	−0.374	x_2^2	−0.090	−0.048
x_1x_2	−0.480	−0.233	x_3^2	−1.001	−0.027
x_1x_3	−2.034	−0.068			

- $\hat{\mathbf{b}}'\hat{\mathbf{b}}$: 6.77 for least squares, 0.55 for ridge ($\theta = 0.032$).
- The large least-squares coefficients of x_1x_3, x_1^2 and x_3^2 offset one another along the near dependency; ridge removes most of them.

Ridge Regression in R

- By hand: `solve(crossprod(W) + theta * diag(ncol(W)), crossprod(W, ys))`, with `W` and `ys` scaled to unit length.
- `MASS::lm.ridge(y ~ ., data, lambda = n * theta)`: `lm.ridge` scales the regressors by their standard deviation with divisor n, so its `lambda` is n times our θ. `coef()` returns coefficients in the original units.
- `glmnet::glmnet(X, y, alpha = 0)` uses yet another scaling of the penalty, so its `lambda` is not comparable with θ.

Always check which scaling a package uses before comparing values of θ.

Principal-Component Regression

Rotate the regressors onto the eigenvectors: $\mathbf{Z} = \mathbf{WV}$, $\boldsymbol{\alpha} = \mathbf{V}'\mathbf{b}$.

$$\mathbf{y}^* = \mathbf{Wb} + \boldsymbol{\varepsilon}^* = \mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}^*, \qquad \mathbf{Z}'\mathbf{Z} = \mathbf{V}'\mathbf{W}'\mathbf{WV} = \boldsymbol{\Lambda}.$$

- The columns of $\mathbf{Z}$ (the principal components) are orthogonal, so $\hat{\alpha}_j = \mathbf{z}_j'\mathbf{y}^*/\lambda_j$ and $\mathrm{Var}(\hat{\alpha}_j) = \sigma_w^2/\lambda_j$.
- Components with small λ_j have huge variance. PCR sets their α_j to 0 and keeps the first m:

$$\hat{\mathbf{b}}_{PC} = \sum_{j=1}^{m} \hat{\alpha}_j \mathbf{v}_j.$$

- Variance $\sigma_w^2 \sum_{j \le m} 1/\lambda_j$; bias from the dropped components, unless their true α_j are 0.

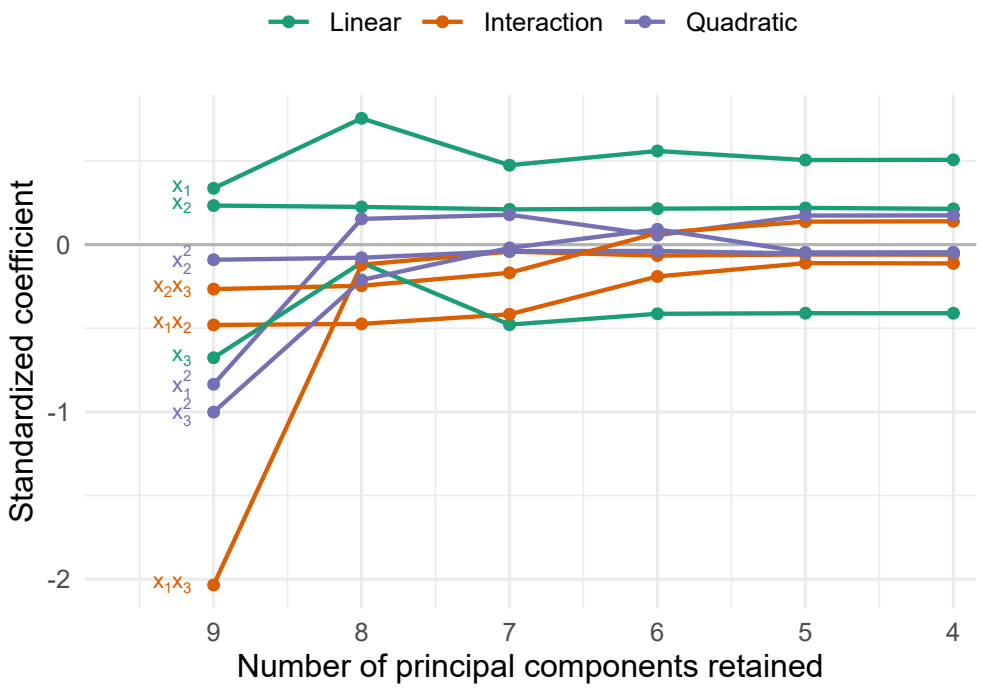
Acetylene Data: Principal Components

j	1	2	3	4	5	6	7	8	9
λ_j	4.21	2.16	1.14	1.04	0.385	0.050	0.014	0.005	0.0001
$\hat{\alpha}_j$	−0.352	0.005	0.602	−0.234	0.009	0.217	0.383	0.520	−2.401
$\hat{\alpha}_j/\,\mathrm{se}(\hat{\alpha}_j)$	−36.9	0.4	32.8	−12.2	0.3	2.5	2.3	1.9	−1.2

- $\hat{\alpha}_9 = -2.40$ is the largest estimate but also the least reliable: its standard error is $\sqrt{\hat{\sigma}_w^2/\lambda_9} = 1.99$.
- Drop components 7–9 ($\kappa_j > 100$) and keep $m = 6$: $R^2 = 0.9938$, $\mathrm{SS}_{\mathrm{Res}} = 13.26$, all VIFs below 8.3.
- PCR chooses components by their eigenvalue, not by how well they predict y. A component with small λ_j can still matter for y.

Acetylene Data: PCR Coefficients

- $m = 9$ reproduces least squares.
- Dropping the 9th component alone already brings the coefficient of x_1x_3 from −2.03 to −0.12.
- From $m = 6$ downward the coefficients barely move.



Acetylene Data: Three Estimates Side by Side

	LS	Ridge	PCR
x_1	0.337	0.539	0.560
x_2	0.234	0.212	0.215
x_3	−0.676	−0.374	−0.413
x_1x_2	−0.480	−0.233	−0.190
x_1x_3	−2.034	−0.068	−0.065
x_2x_3	−0.266	0.012	0.069
x_1^2	−0.835	0.125	0.056
x_2^2	−0.090	−0.048	−0.038
x_3^2	−1.001	−0.027	0.092
R^2	0.9977	0.9944	0.9938
Largest VIF	6563	4.18	8.27

Least squares (LS), ridge with $\theta = 0.032$, PCR with $m = 6$ components.

- Ridge and PCR agree closely with each other.
- Both differ sharply from least squares on the temperature and contact time terms.
- Both give up less than 0.004 in R^2.

Other Biased Estimators

- **Lasso**: penalty $\theta \sum_j |b_j|$ instead of $\theta \sum_j b_j^2$. It shrinks and also sets some coefficients exactly to 0, so it selects regressors as well.
- **Elastic net**: a mix of the ridge and lasso penalties; useful when groups of correlated regressors should enter together.
- **Latent root regression** (Webster, Gunst and Mason, 1974): like PCR, but uses the eigenvectors of the correlation matrix of y and the regressors together.

All trade a little bias for a large reduction in variance.

Question: Why Accept a Worse Fit?

- Ridge and PCR always have a larger $\mathrm{SS}_{\mathrm{Res}}$ than least squares on the same data. Why would we prefer them?
- Should the regressors be scaled before ridge regression? Does it matter?

8 Summary

Diagnostics at a Glance

Diagnostic	Warning level	What it shows
Pairwise r_{jl}	$\lvert r_{jl}\rvert$ near 1	which pairs (misses larger groups)
$\mathrm{VIF}_j = 1/(1 - R_j^2)$	above 5 or 10	which coefficients suffer
$\kappa = \lambda_{\max}/\lambda_{\min}$	100–1000; above 1000 severe	overall severity
κ_j	above 1000	how many dependencies
Eigenvector $\mathbf{v}_j$	large loadings	which regressors are involved
Proportions π_{sj}	two or more above 0.5	which coefficients each affects

The Examples at a Glance

Data	Source	Largest VIF	κ	Remedy shown
Acetylene	sampling, model terms	6563	43,381	ridge, PCR
Webster	constraint	298	2196	—
Gasoline (10 reg.)	population	161	1820	—
Hald cement	constraint	283	1377	drop x_4
Windmill	model terms	38.5	152	centering
Delivery time	— (mild)	3.12	10.4	none needed

Key Takeaways

- Multicollinearity is a near linear dependence among the regressors. Least squares stays unbiased, but its variances can be enormous.
- The data pin down combinations of correlated coefficients well and the individual coefficients poorly; prediction inside the data region is often fine.
- Pairwise correlations can miss a dependency; VIFs and the eigenvalues of $\mathbf{W}'\mathbf{W}$ do not.
- Eigenvectors and variance-decomposition proportions show which regressors are involved.
- Remedies: better data, a better parameterization, or a biased estimator such as ridge regression or PCR that trades a little bias for a large drop in variance.